Homework 4
11 February 2025
Total Pages: 5

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Question 1

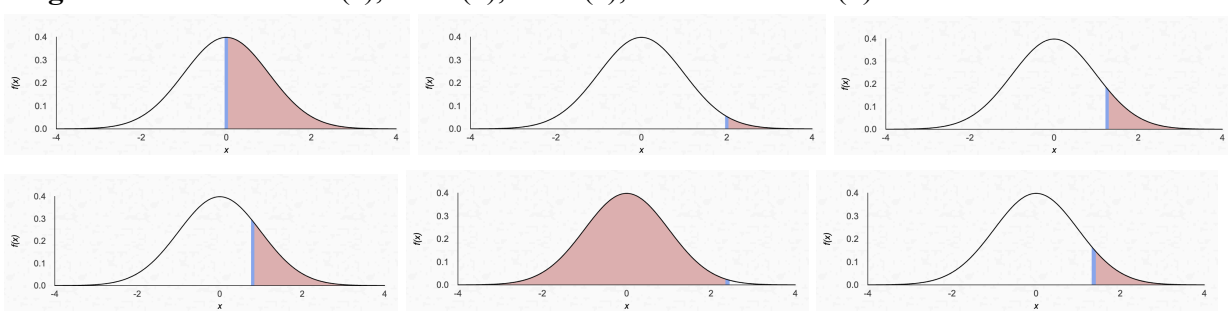
Given:
Intervals

a. Between $z = 0$ and $z = 2.0$
b. To the right of $z = 1.26$
c. To the left of $z = -0.8$
d. Between $z = -1.47$ and $z = 2.39$

ZtableNegativeInfinity.pdf

Find:
Find the area under the standard normal distribution curve for the above intervals.

Diagram: 1 and 2 for (a), 3 for (b), 4 for (c), and 5 and 6 for (d)



Theory:

Table ZtableNegativeInfinity.pdf has the P values of Z ≥ 0.

Assumptions:
Normal distribution, independent events, continuity of the distribution, and use of the CDF function.

Solution:

a) $P(0 \leq Z \leq 2) = P(Z \leq 2) - P(Z \leq 0) = 0.9772 - 0.5 = 0.4772$
b) $P(Z > 1.26) = 1 - P(Z \leq 1.26) = 1 - 0.8962 = 0.1028$
c) $P(Z \leq -0.8) = 0.2119$
d) $P(-1.47 \leq Z \leq 2.39) = P(Z \leq 2.39) - P(Z \leq -1.47) = 0.9916 - 0.0708 = 0.9208$

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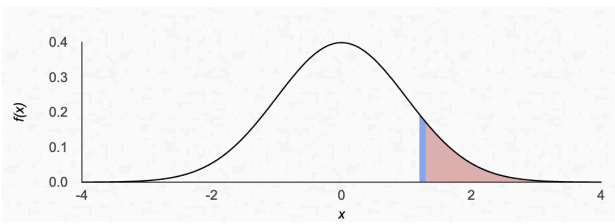
Question 2

Given:
Quiz grades normally distributed with a mean of 85 and a standard deviation of 8.0.
ZtableNegativeInfinity.pdf

Find:
The probability that a student will have a quiz grade of 95 or greater.

Diagram:



Theory:
Z-score: $Z = \frac{X-\mu}{\sigma}$
Table ZtableNegativeInfinity.pdf has the P values of Z ≥ 0.

Assumptions:
Normal distribution, independent events, continuity of the distribution, and use of the CDF function.

Solution:
Get the Z-score: $Z = \frac{X-\mu}{\sigma} = \frac{95-85}{8} = 1.25$

Find the probability
$P(Z \geq 1.25) = 1 - P(Z \leq 1.25) = 1 - 0.8944 = 0.1056$

As a percent: **10.56%**

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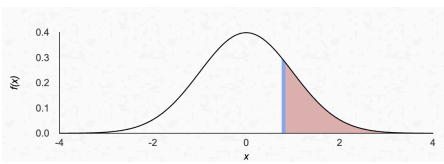
Question 3

Given:
The amount of soda in the machine, normally distributed with a mean of 63.0 oz and a standard deviation of 1.25 oz.
ZtableNegativeInfinity.pdf

Find:
The probability that a randomly selected cup will be overfilled if each cup holds 64 oz?

Diagram:



Theory:
Z-score: $Z = \frac{X-\mu}{\sigma}$
Table ZtableNegativeInfinity.pdf has the P values of Z ≥ 0.

Assumptions:
Normal distribution, independent events, continuity of the distribution, and use of the CDF function.

Solution:
Get the Z-score: $Z = \frac{X-\mu}{\sigma} = \frac{64-63}{1.25} = 0.8$

Find the probability
$P(Z \geq 0.8) = 1 - P(Z \leq 0.8) = 1 - 0.7881 = 0.2119$

As a percent: **21.19%**

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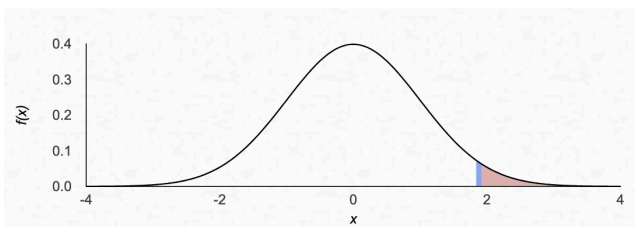
Question 4

Given:
A company producing light bulbs that last 4600 hours with a standard deviation of 250 hrs.
ZtableNegativeInfinity.pdf

Find:
The lifetime the company should promote for these bulbs so only 3.0% of them burn out before.

Diagram:



Theory:
We can get Z from P using scipy norm.ppf
Reverse the Z-score formula, $Z = \frac{X-\mu}{\sigma}$, becomes $X = \mu + \sigma Z$

Assumptions:
Normal distribution, independent events, continuity of the distribution, and use of the PPF function.

Solution:
Get the Z score from P $\quad Z = norm.ppf(0.03) = -1.881$

The the claimed lifetime $\quad X = \mu + \sigma Z = 4600 + 250(-1.881) = 4130$

The claimed lifetime should be **4130**

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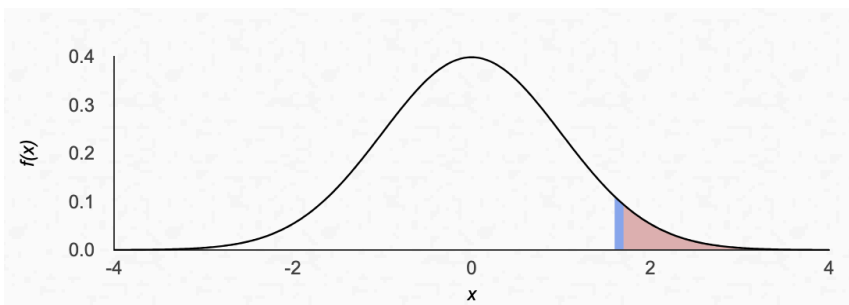
Question 5

Given:
The life of an eclectic scooter being 13 years with a standard deviation of 1.5 years.
ZtableNegativeInfinity.pdf

Find:
The time of the warranty such that only 5% fail while under warranty.

Diagram:



Theory:
We can get Z from P using scipy norm.ppf
Reverse the Z-score formula, $Z = \frac{X-\mu}{\sigma}$, becomes $X = \mu + \sigma Z$

Assumptions:
Normal distribution, independent events, continuity of the distribution, and use of the PPF function.

Solution:
Get the Z score from P $\quad Z = norm.ppf(0.05) = -1.645$

The the claimed lifetime $\quad X = \mu + \sigma Z = 13 + 1.5(-1.645) = 10.53$

The warranty time should be **10.53 years**